Paving By The Numbers



Important Safety Information

Most accidents involving product operation, maintenance and repair are caused by failure to observe basic safety rules and precautions. An accident can often be avoided by recognizing potentially hazardous situations before an accident occurs. A person must be alert to potential hazards. This person should also have the necessary training, skills and tools to perform these functions properly.

Improper operation, lubrication, maintenance or repair of this product can be dangerous and could result in injury or death.

Do not operate or perform any lubrication, maintenance or repair on this product, until you have read and understood the operation, lubrication, maintenance and repair information.

The hazards are identified by the “Safety Alert Symbol” and followed by a “Signal Word” such as “WARNING” as shown below.

⚠ WARNING

The meaning of this safety alert symbol is as follows:
Attention! Become Alert! Your Safety is Involved.

The message that appears under the warning, explaining the hazard, can be either written or pictorially presented.

Operation that may cause product damage are identified by NOTICE labels on the product and in this presentation.

Caterpillar cannot anticipate every possible circumstance that might involve a potential hazard. The warnings in this presentation and on the product are therefore not all inclusive. If a tool, procedure, work method or operating technique not specifically recommended by Caterpillar is used, you must satisfy yourself that it is safe for you and others. You should also ensure that the product will not be damaged or made usafe by the operation, lubrication, maintenance or repair procedures you choose.

The information, specifications, and illustrations in this presentation are on the basis of information available at the time it was written. The specifications, illustrations and other items can change at any time. These changes can effect the service given to the product. obtain the complete and most current informaiton available.

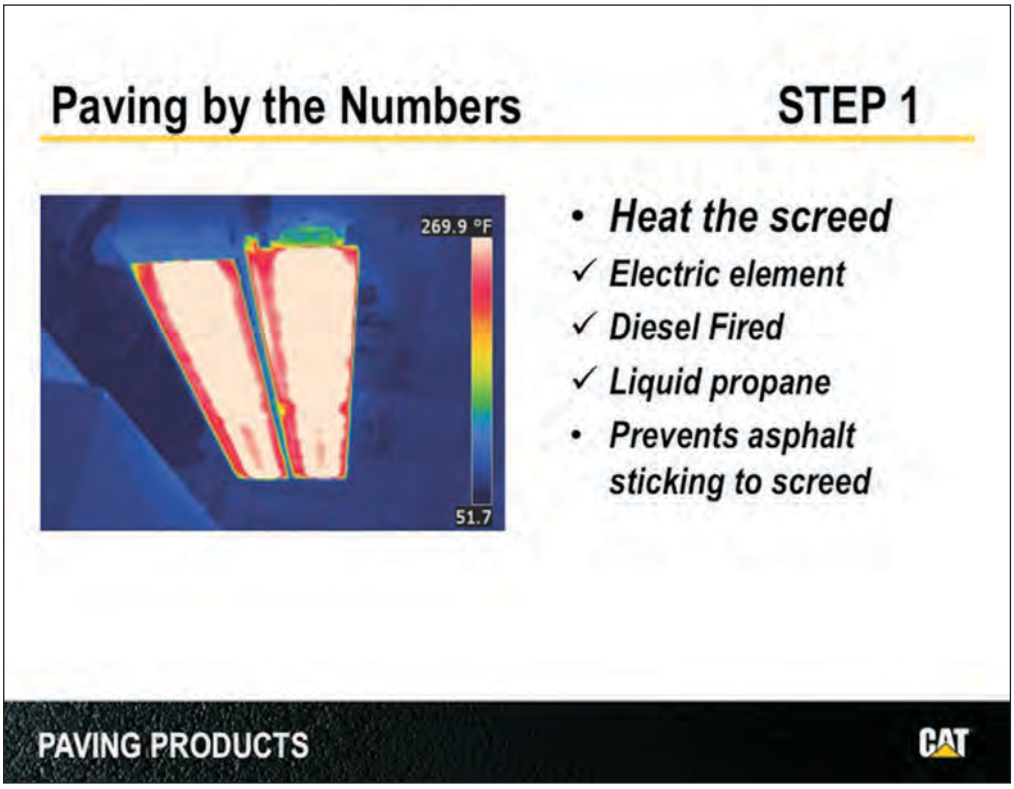
⚠ WARNING

Do not operate or work on a machine unless you have read and understand the instructions and warnings in the Operator and Maintenance Manuals. Failure to follow the instructions or heed the warnings could result in injury or death. Contact any Cat® dealer for replacement manauls. Proper care is your responsibility.



Producing a smooth asphalt surface with a uniform finish is not as simple as saying “one—two—three.” But a well-trained paver crew, which follows a set of established fundamental practices, will avoid many of the problems that affect mat quality.

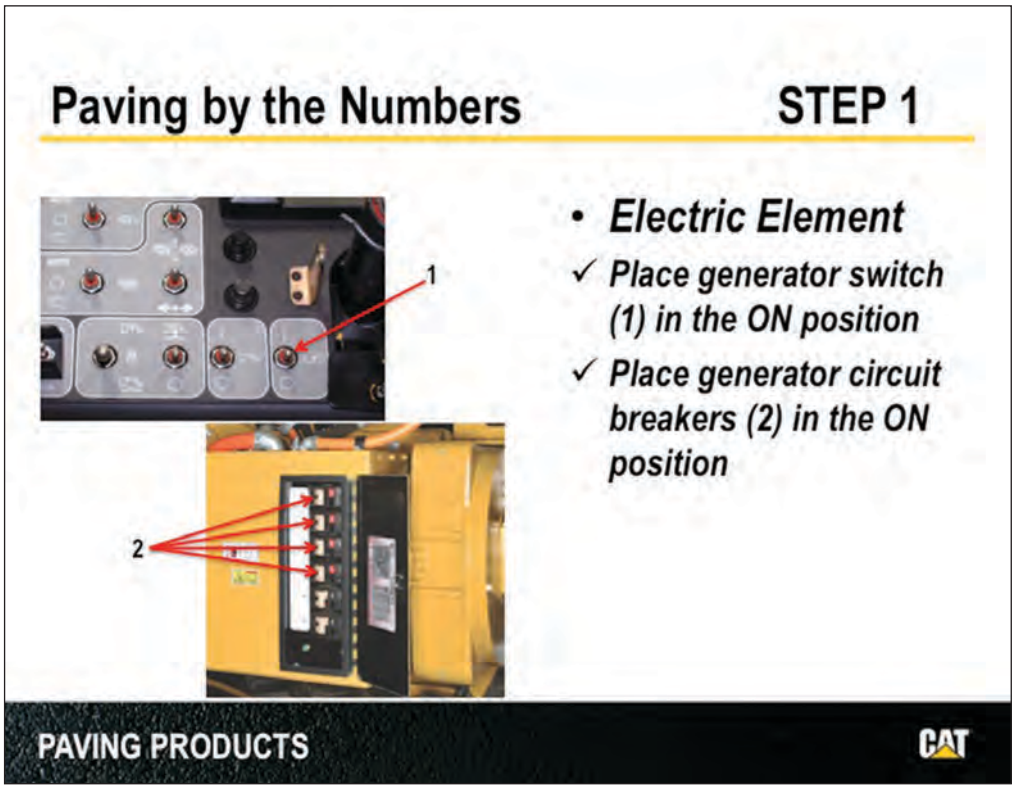
The purpose of this presentation is to review the steps you should follow each time you get ready to pull off the joint and start paving.



STEP 1

Step 1 is heat the screed plates. If the screed is cold, you have to heat it before beginning to pave to prevent asphalt sticking to the screed plates. There are several types of screed heaters on Cat screeds, diesel fired, liquid propane and electric element. In this presentation, we will talk about the electric screed heat only. If you need information on the Diesel or Liquid Propane screed heat, please refer to the Operation & Maintenance Manual (O&MM) for that screed.

On Cat extendable-type screeds, each screed frame section is equipped with a heater. One on each screed extender and two on the main screed.



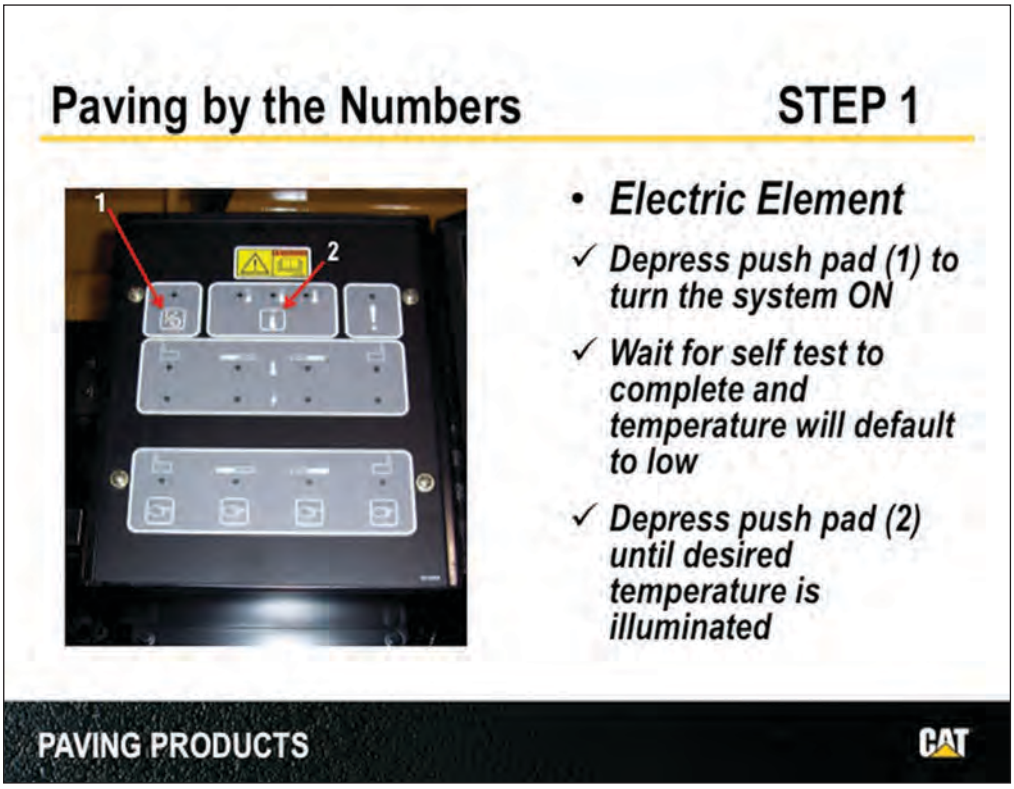
STEP 1 *(continued)*

Electric Element

Locate the generator switch and move it to the ON position. On Cat machines, depending on the machine model, this switch will be located on the left console (shown) or on the center console.

Located on the generator are the circuit breakers for each section of the screed. Move them all to the ON position.

Start with the screed approximately 25-76 mm (1-3”) off of the ground. If the screed is at maximum height, wind can effect the performance of the heaters. Also if the screed is on the ground, the ground acts as a heat sink and it will take longer for the screed to come up to temperature.

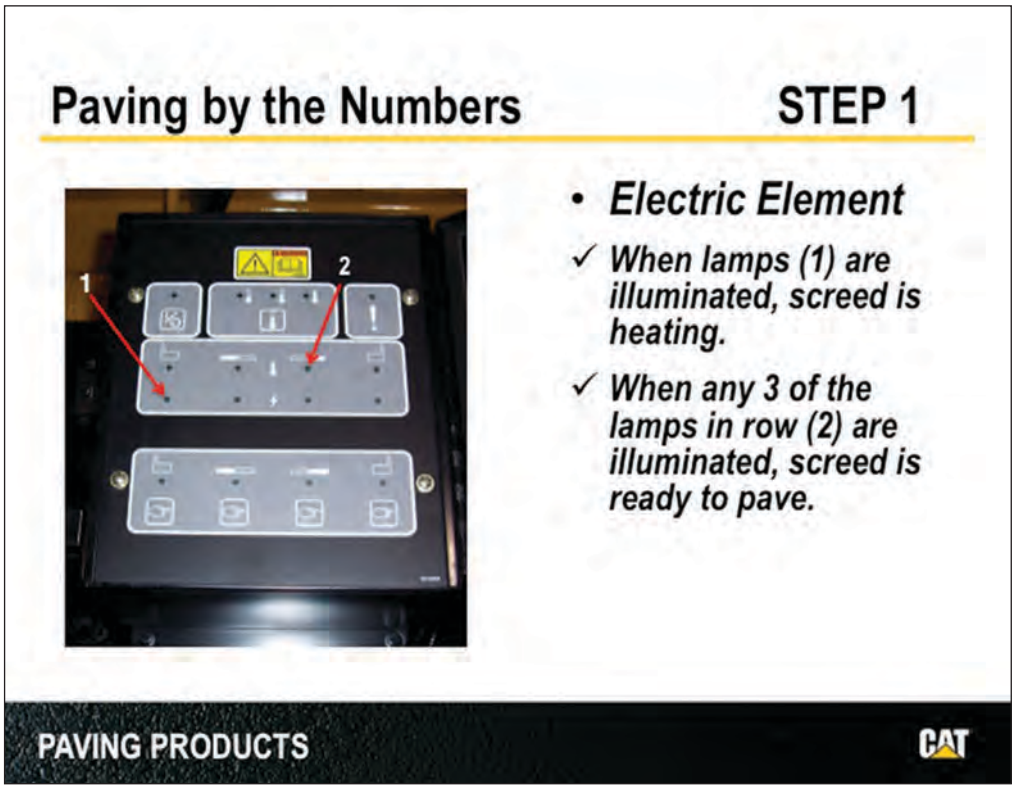


STEP 1 *(continued)*

Depress the push pad (1) to turn the system on. The system will go through a self test where all the lamps will flash several times independently and then all together.

Once the self test is complete, the system will default to the Low temperature setting.

If you want to change the setting to Medium or High, depress the push pad (2) until the desired temperature is illuminated.

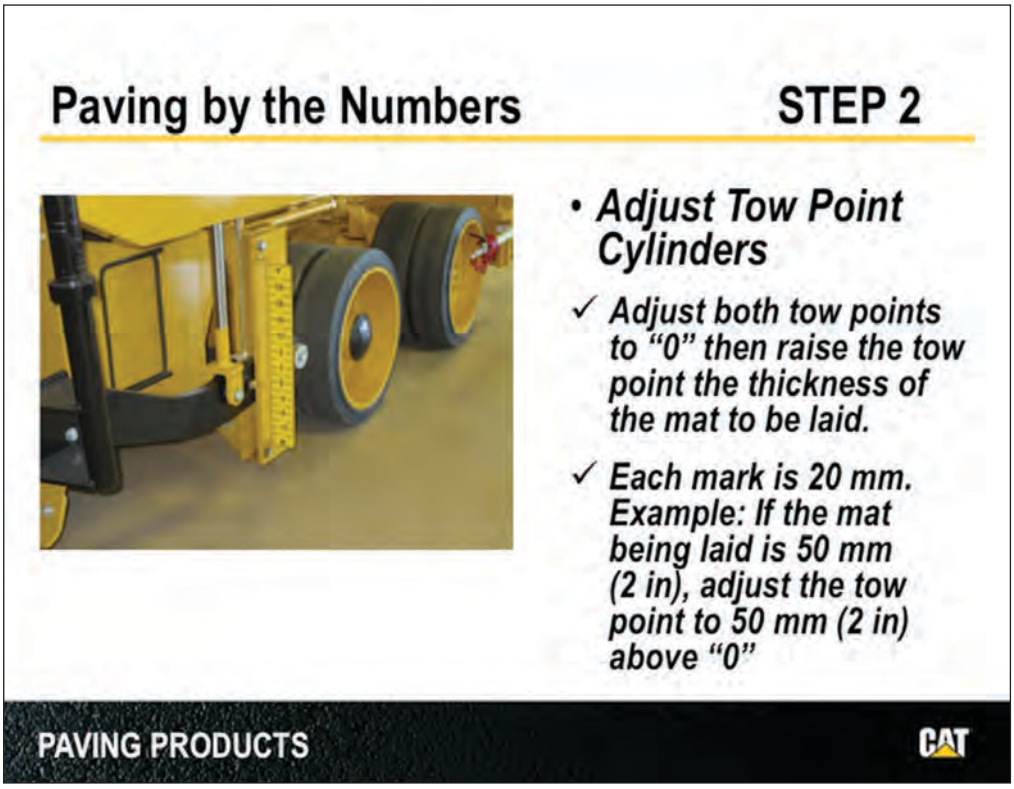


STEP 1 *(continued)*

When an element is powered, the corresponding screed lamp will illuminate (1).

When the screed has reached the set point temperature that was selected, the ready lamp (2) will illuminate.

When any 3 of the screed sections are up to the set point temperature, the remaining section is very close to the set point so paving can begin.

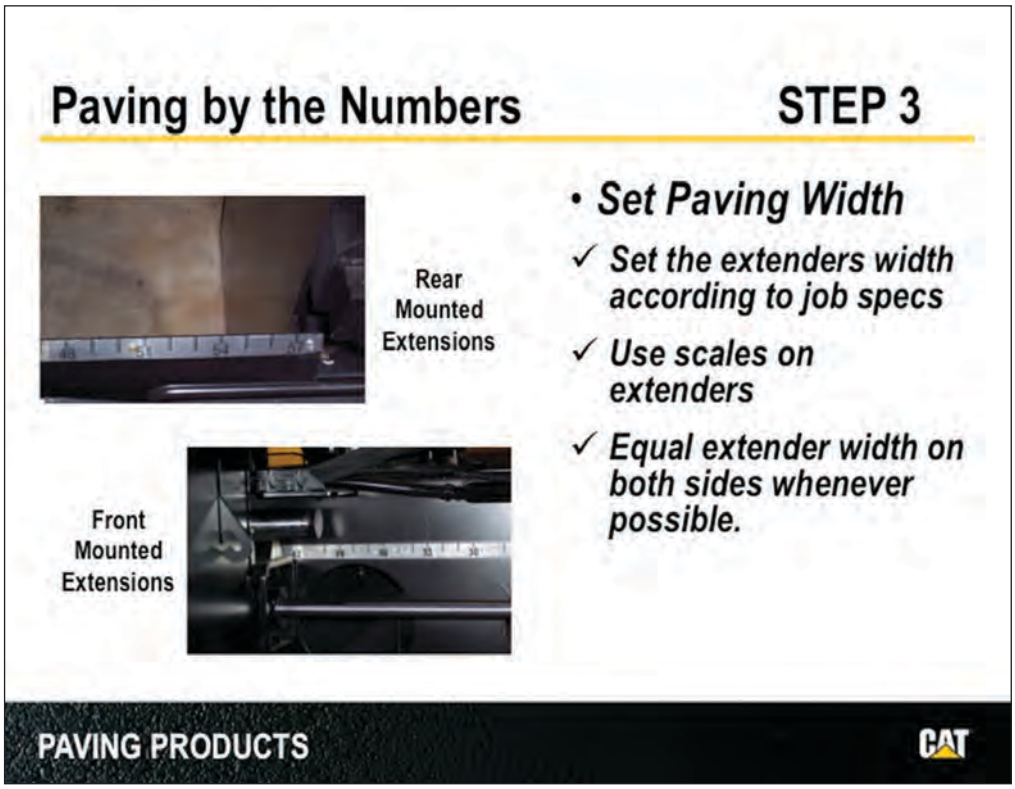


STEP 2

Step 2 is adjusting the tow point cylinder.

After the screed has warmed up, adjust the tow point cylinder on both sides of the machine.

Tow point scales differ depending on screed type but regardless of the type of scale you have each mark indicates 2 cm (3/4 in). On a vibratory screed (AS3301C, AS3251C, AS2252C, AS2302C) the "0" is 14 cm above the lowest mark on the scale. Move the tow point cylinder to "0" then raise the tow point the thickness of the mat being laid. For example, if the mat being laid is 5 cm (2 in), move the tow point up 5 cm (2 in) above "0".

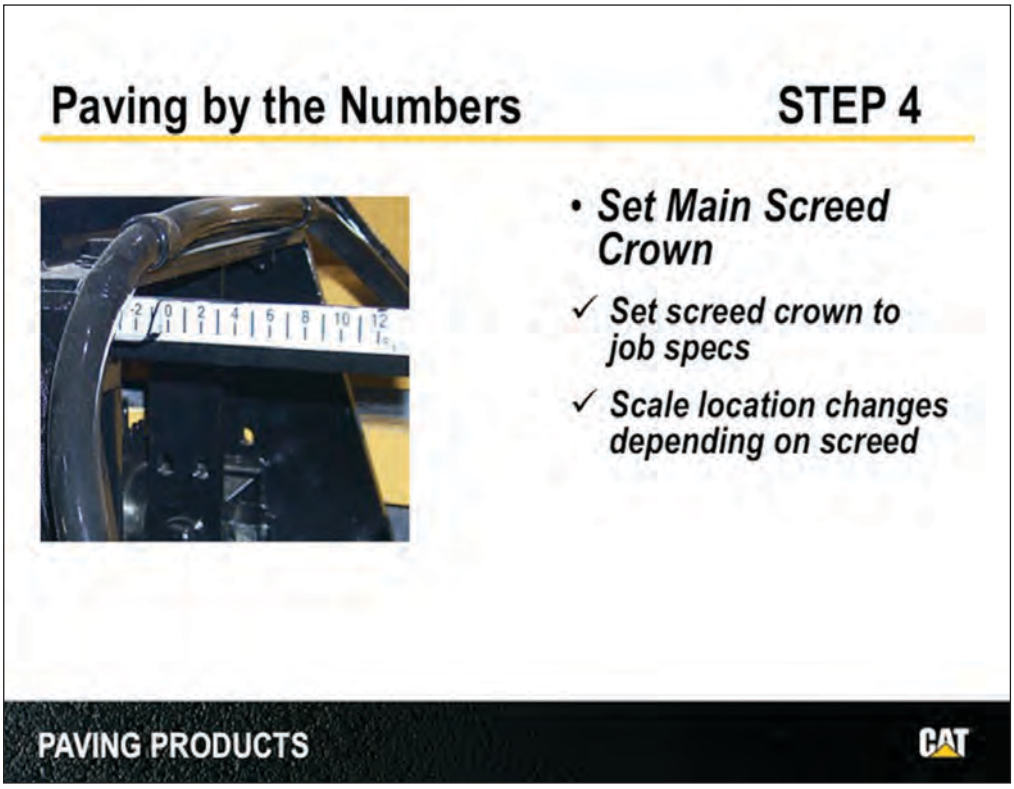


STEP 3

Step 3 is setting the desired paving width.

With the screed still raised, move the extenders out to the specified paving width. Each extender has a scale which shows how far the extender is moved out from the main screed.

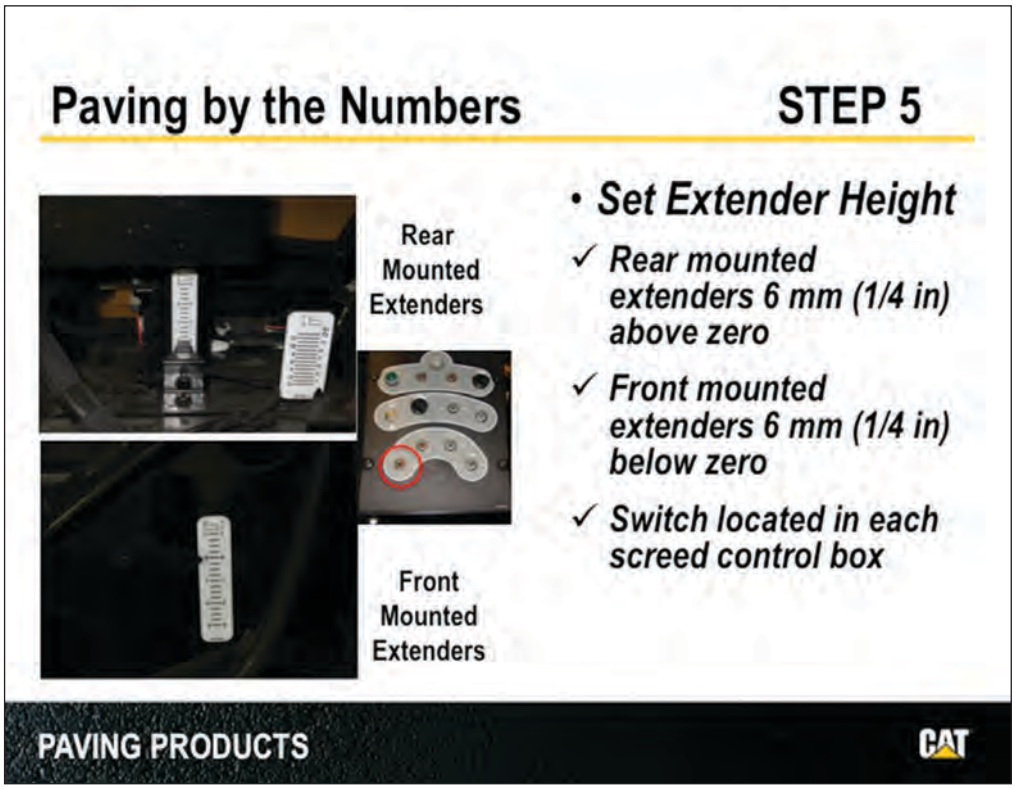
It's good paving practice to balance the extender width so the forces against each extender are equalized. For example, if you need a total of 1.2 m (4 ft) of extender, move the left and right extenders out an equal amount of .6 m (2 ft.) on each side.



STEP 4

Step 4 is setting the main screed crown to the specifications of the job.

Use the crown indicator scale located in the center of the screed to set the crown. Before using the indicator scale, be sure that it is properly calibrated to the screed. The crown indicator scale is located in different locations depending on the type of screed you have.



Step 5

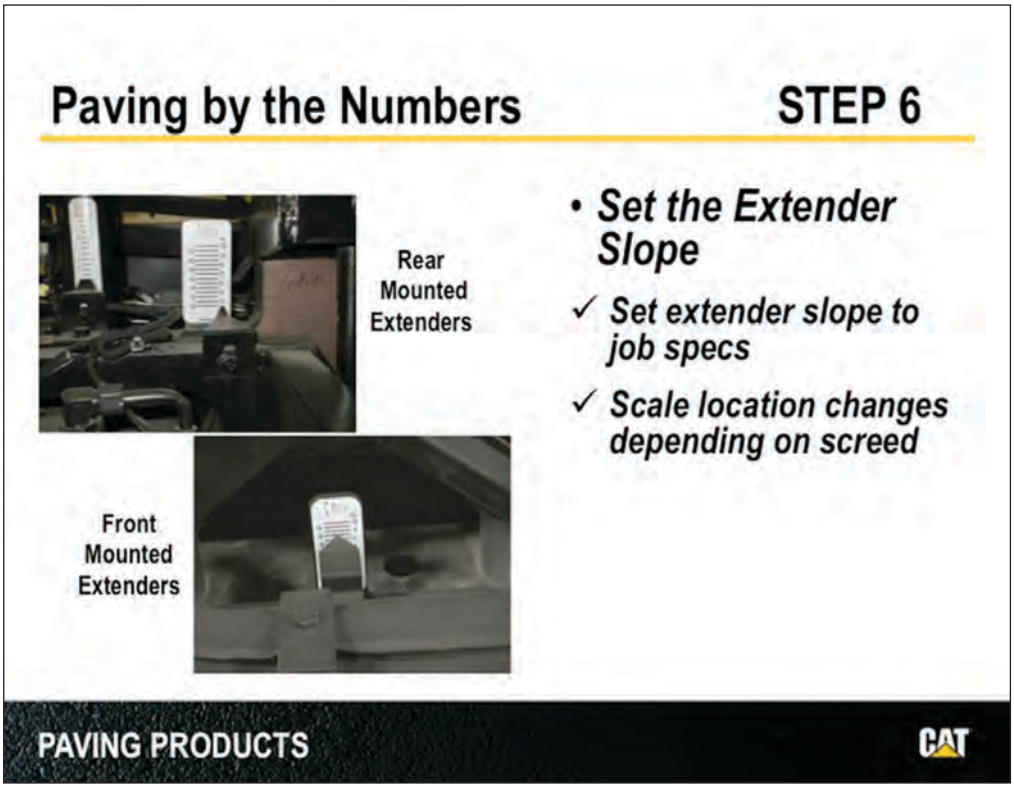
Step 5 is setting the height of the extenders.

You want to set the height of the extenders so they match the height of the main screed. When the extender height is right, there are no transition marks in the mat.

Extender height is adjusted with a switch in each screed control box on vibratory screeds.

Screeds normally run with a 6 mm (1/4 in) nose-up angle of attack. Therefore, on a screed with rear mounted extenders, set the extender height 6 mm (1/4 in) above the zero point on the scale. Do this for both left and right extenders. If the screed has front mounted extenders, lower both extenders 6 mm (1/4 in) below the zero point on the scale.

Again, make sure that the height indicators have been calibrated to the extenders.

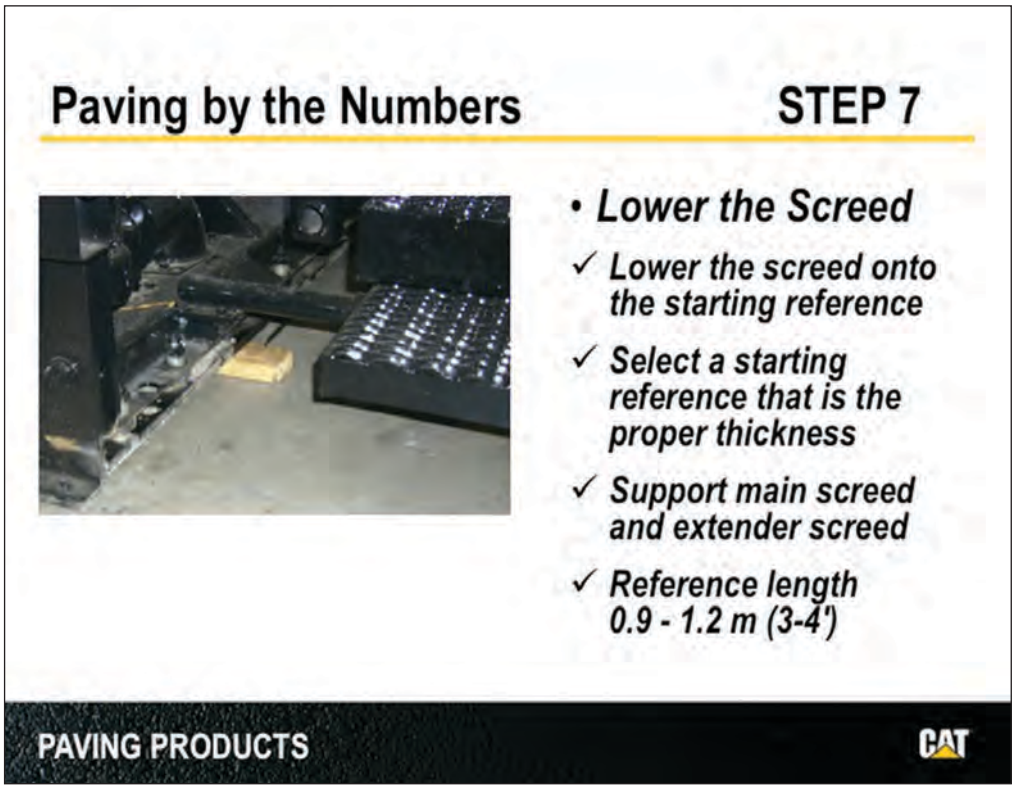


STEP 6

Step 6 is setting the slope of the extenders to the specifications of the job.

If there is no requirement for extender slope, set the indicator at zero on both sides. Location of slope indicator changes depending on screed type.

Again, make sure that the Slope indicators have been calibrated to the extenders.



STEP 7

Step 7 is positioning the starting references under the screed.

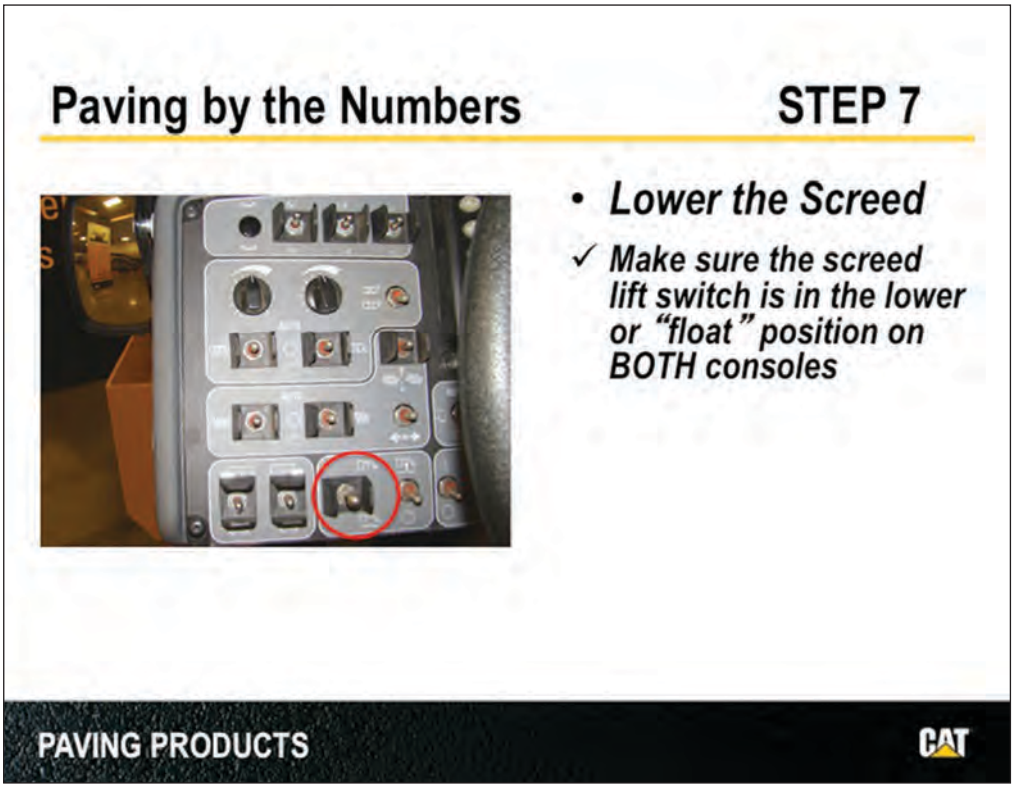
Select a starting reference whose thickness matches the mat depth and the rate of compaction. Normally, the mat will compact about 6 mm (1/4 in per inch) per 25 mm of mat depth. For example, to get a 50 mm (2 in) compacted mat, use a starting reference which is 64 mm (21/2 in) thick.

Use two references. Position them so they completely support both the main screed and the screed extender from front to back. The normal length is between 0.9 - 1.2 m (3-4 ft).

When the starting references are in position, lower the screed so it rests on both references.

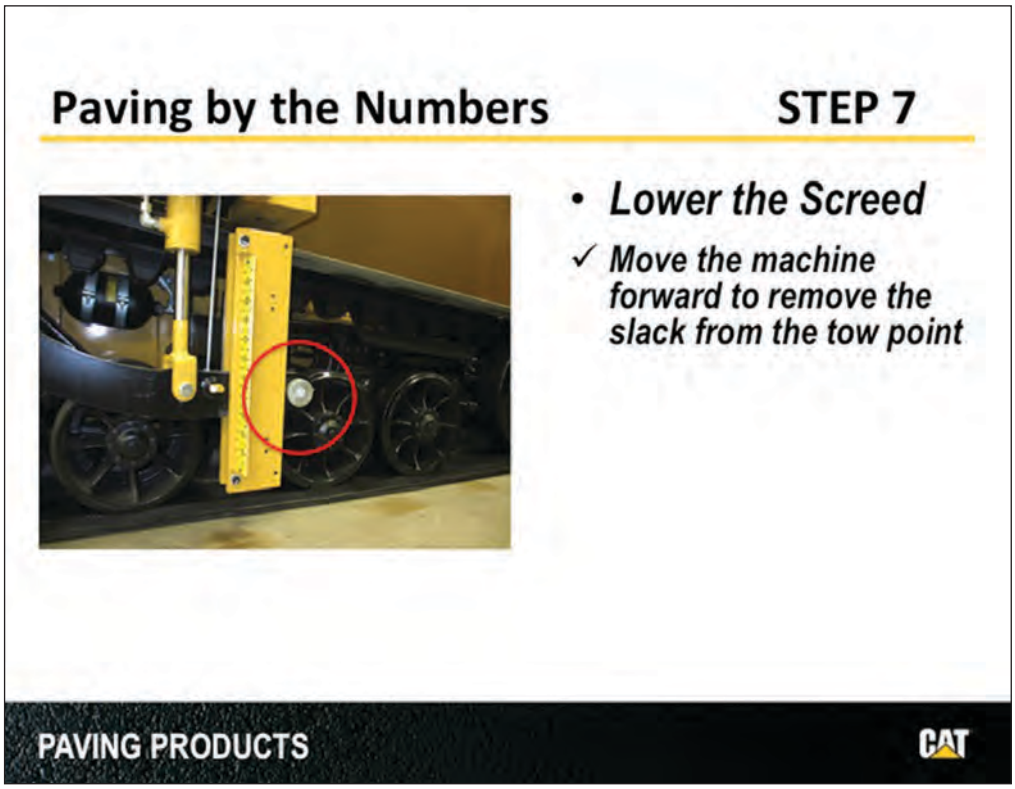
NOTE: When paving with the extenders beyond the halfway point, an additional starting reference for the extender may be required.

NOTE: Ensure that the end gates are adjusted so the screed rests on the starting reference and not the end gate.



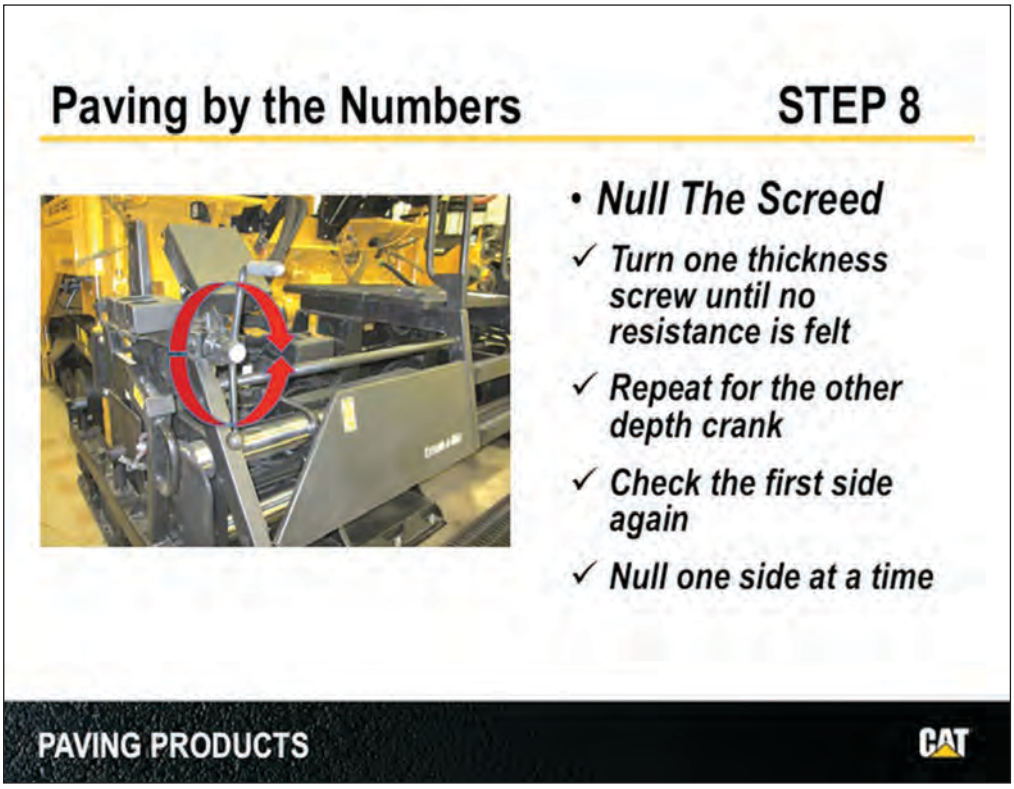
STEP 7 *(Continued)*

Remember to leave the screed RAISE/LOWER switch in the lower or "float" position on BOTH consoles.



STEP 7 *(Continued)*

Step 7 also includes removing the slack from the tow point. Move the machine forward until the tow arm roller contacts the tow point frame.

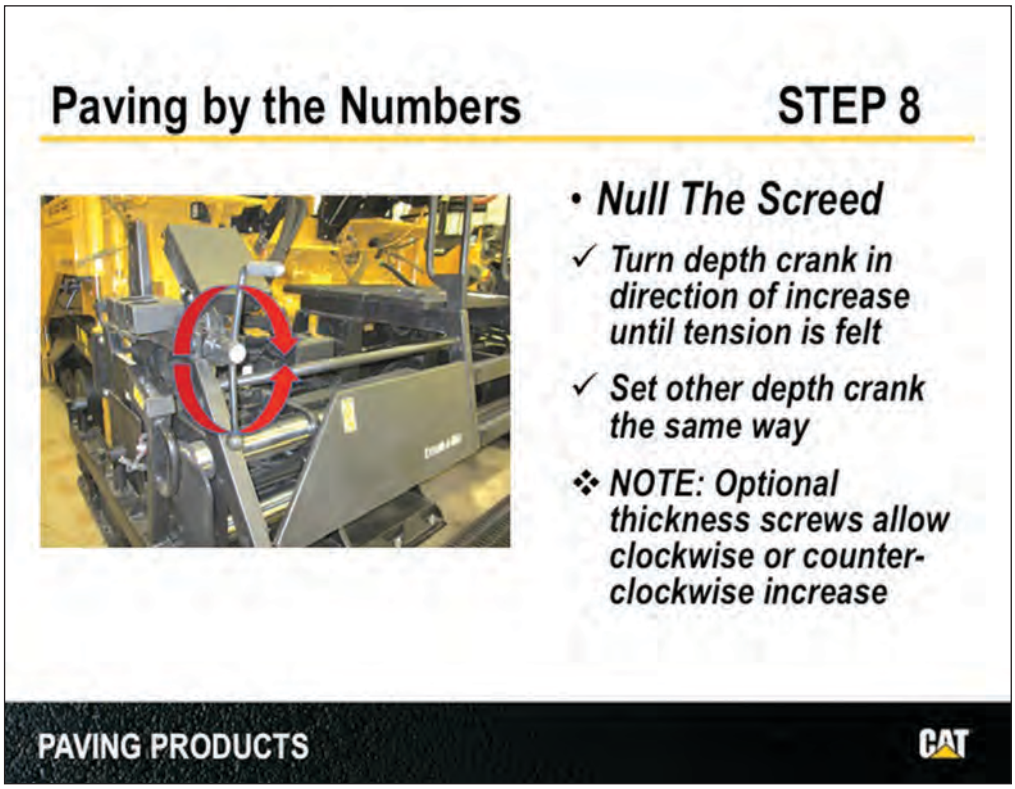


STEP 8

Step 8 is nulling the screed.

Use the manual depth control cranks to null the screed. Turn the crank in either direction until no resistance is felt. This ensures that the full weight of the screed is supported by the starting reference. Move to the other side and follow the same nulling procedure. Be sure the crank turns freely. Then, go back to the other side. Make sure the crank still turns freely.

NOTE: Repeat this until both sides have no resistance.



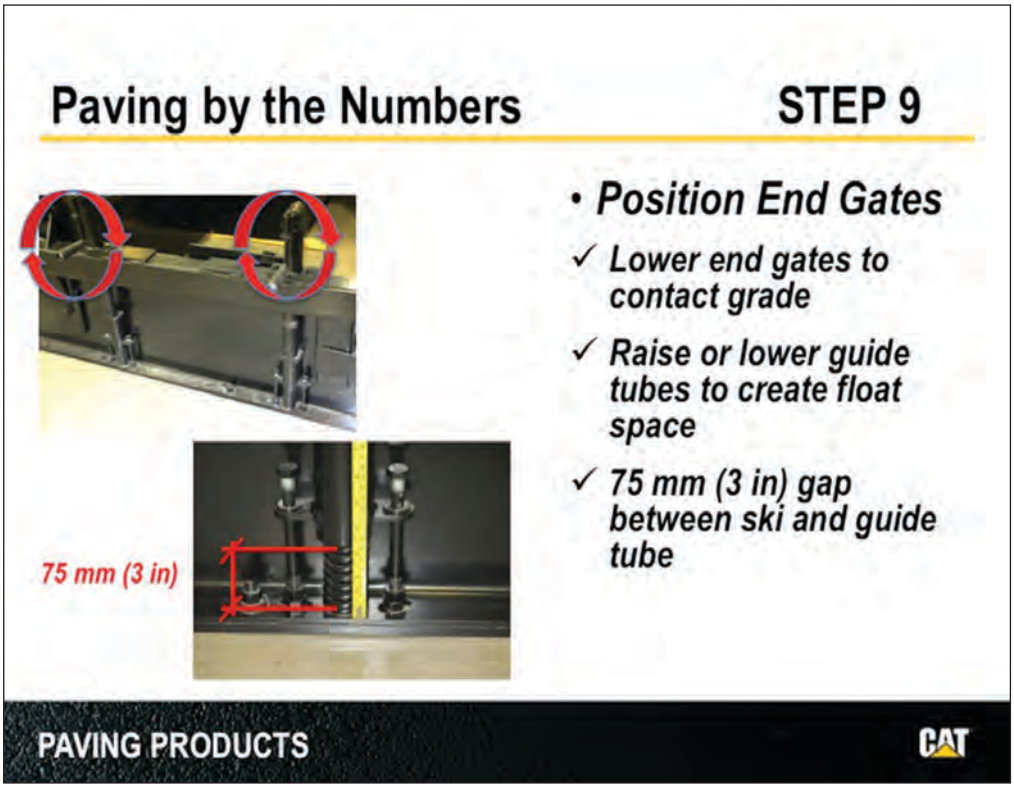
STEP 8 *(Continued)*

Then, turn the depth crank in the direction that increases mat thickness until resistance is felt.

On the other side, turn the crank until you feel resistance.

On other types of screeds, you may have to turn the depth cranks one or more revolutions to set the angle of attack. Follow the manufacturer's guidelines for the screed you're setting up.

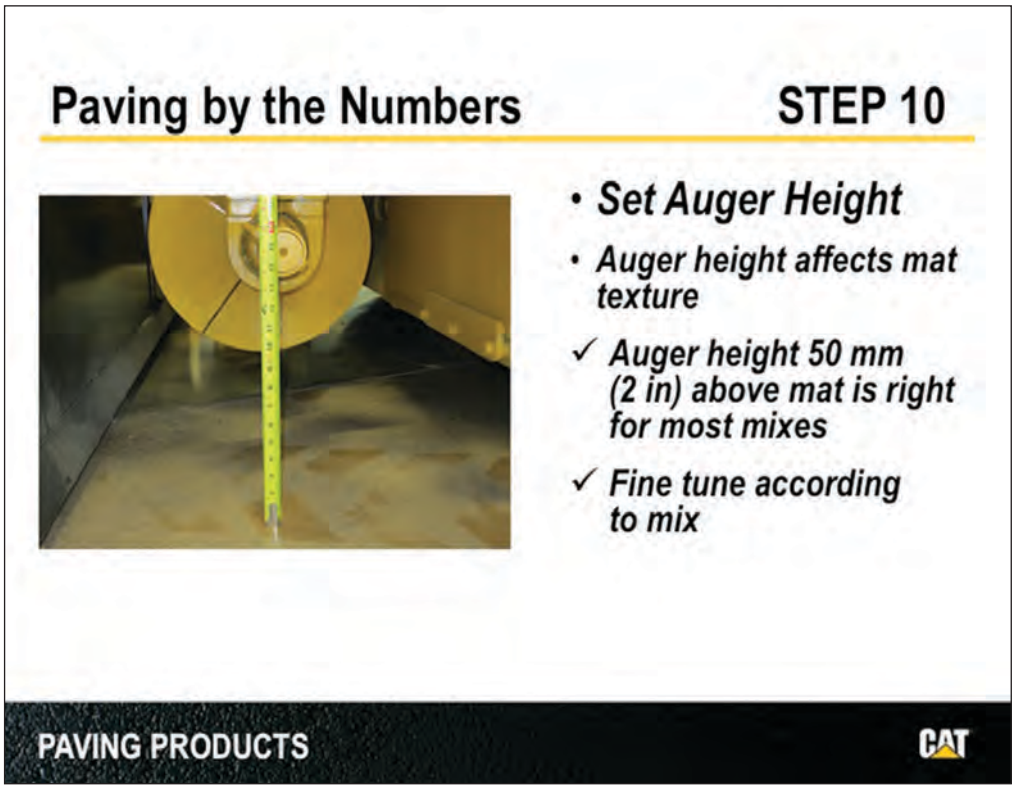
NOTE: Optional thickness screws allow for clockwise or counter clockwise rotation of the screw to increase the angle of attack.



STEP 9

Step 9 is positioning the end gates.

End gates on both ends of the screed retain material at the proper width. They float on the grade. First, lower the end gate until it touches the grade. Next, raise or lower the guide tubes to create space for the spring-loaded skis to move up and down over any grade irregularities. The space should be about 76 mm (3 in).



STEP 10

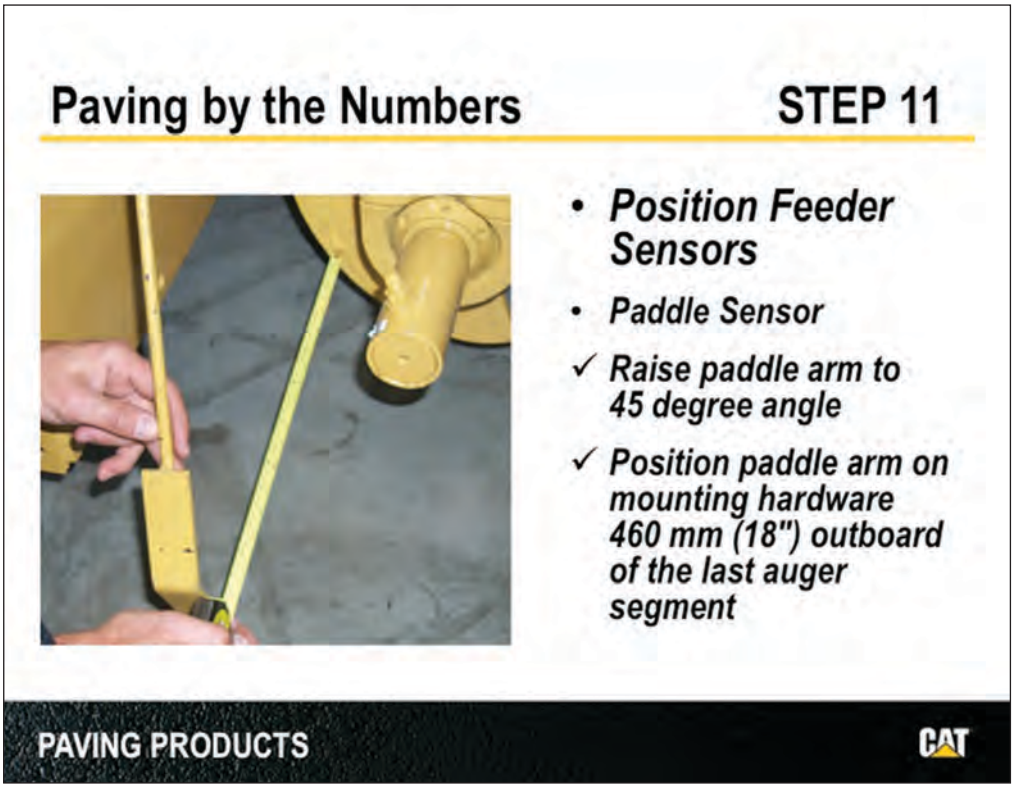
Step 10 is setting the auger height.

The height of the augers in relation to the depth of the uncompacted mat has an effect on the texture and finish of the mat.

If the augers are too low, you'll probably see an open texture and maybe material segregation. If the augers are too high, the head of material will likely be too high which will cause the screed to climb. Different mixes react differently to auger height adjustment, but as a rule, set the auger height at least 5 cm (2 in) above the height of the uncompacted mat.

From the bottom of the auger to the center of the auger shaft is 203 mm (8 in). Now add 50 mm (2 in) to that. Then add the thickness of mat to be paved. The total is the distance from the centerline of the auger shaft to the grade. Adjust the auger up or down until the dimension is reached.

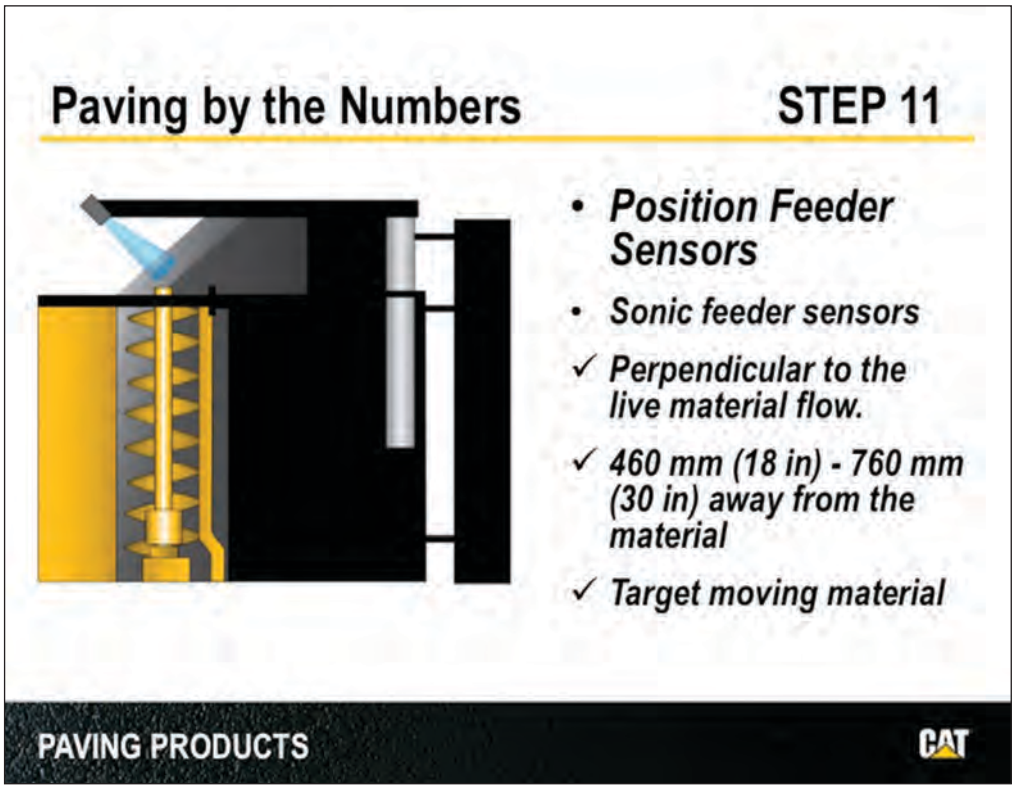
Fine tune the auger height after paving starts if the mix is very coarse or very tender.



STEP 11

Step 11 is setting the position of the feeder sensors.

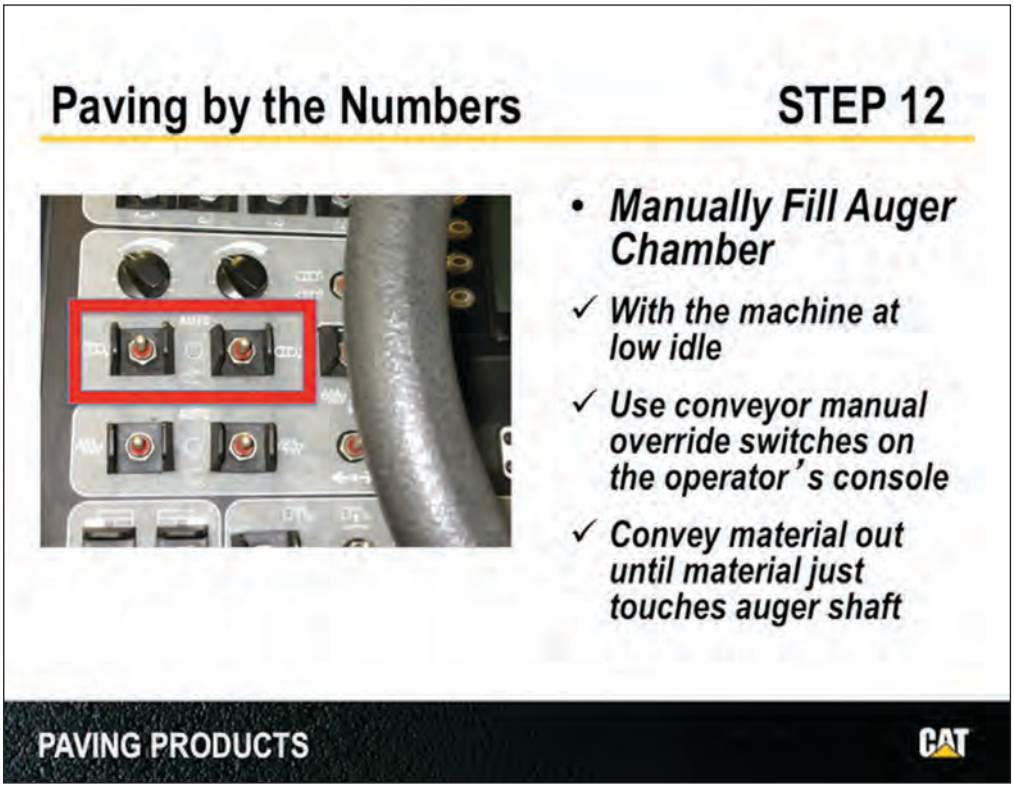
You want to position the paddle type feeder sensors so they are sensing the active pile of material about 460 mm (18 in) away from the last auger segment. With the paddle-type sensor, the paddle arm should be at a 45 degree angle at the 460 mm (18 in) distance.



STEP 11 *(Continued)*

Step 11 is setting the position of the feeder sensors.

Sonic feeder sensors should be set perpendicular to the material flow 460 mm (18 in) - 76 cm (30 in) away from the material. Always target the moving material.



STEP 12

Step 12 is filling the auger chamber in front of the screed until the material is covering one half the auger.

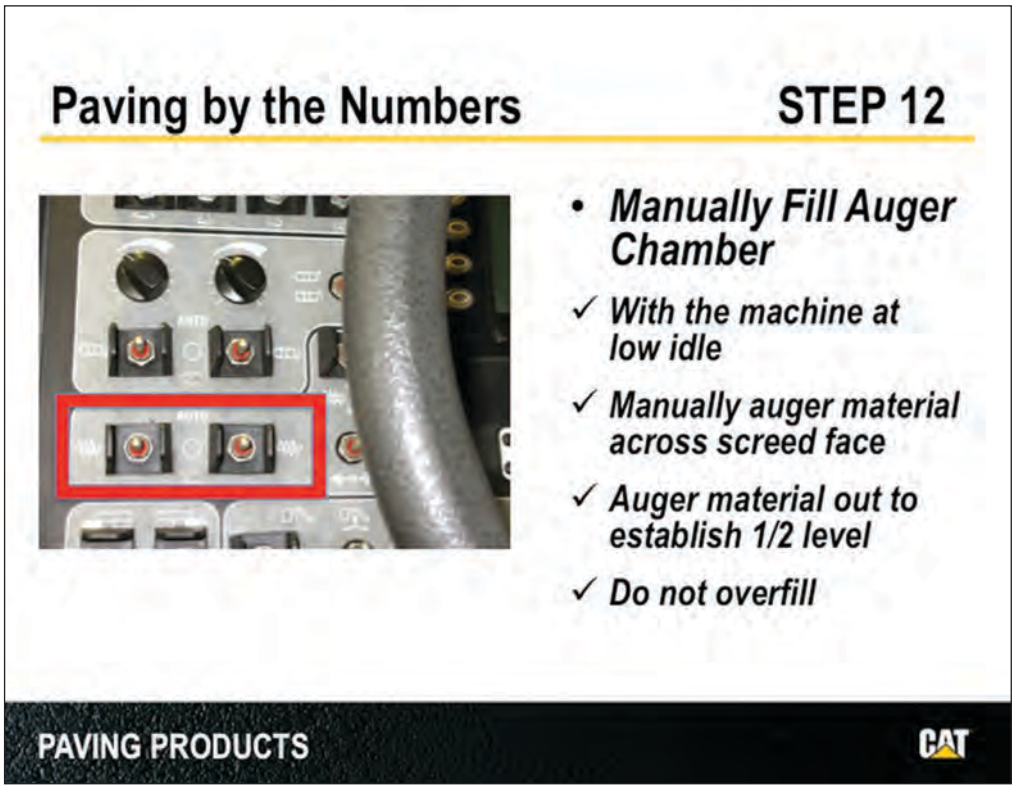
Fill the chamber with the engine at low idle, one side at a time, alternating between the conveyor and auger. This helps prevent material from being pulled under the machine and overfilling the chambers.

NOTE: Do not fill the chamber using the conveyor and auger switches together or by using the screed override switches. This will overfill the chambers and cause a bump on take off.

Using the manual override switches on the operator's console, alternately convey and auger material to the auger chamber and out to the ends of the augers.

The goal is to fill the auger chamber evenly on both sides.

Use one conveyor switch at a time to move material out until it just touches the auger shaft.



STEP 12 *(Continued)*

Then, use the manual auger switch to move material out to the end of the screed. The correct head of material is one that covers one half of the augers.

Always fill the auger chamber using the manual mode at low idle and be careful not to overfill.

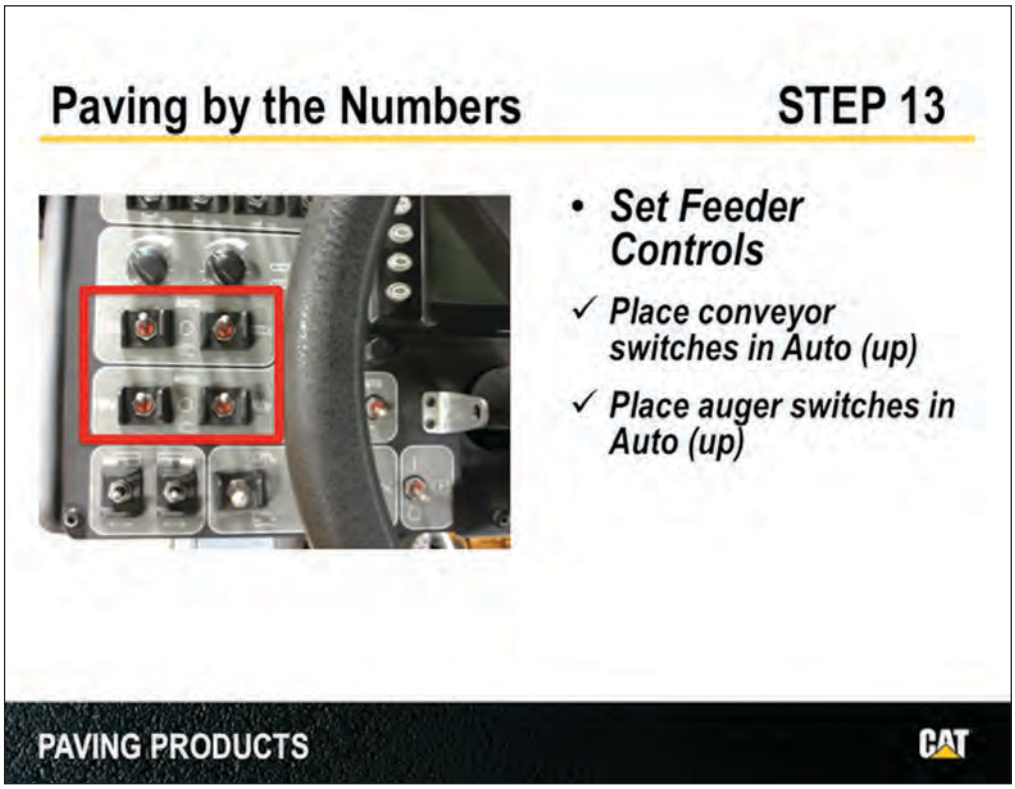
Overfilling the auger chamber will result in a bump when starting off the transverse joint.



STEP 12 *(Continued)*

Don't force material out to the end gate. You'll probably overfill the chamber. Instead, move some material with a shovel to the end gate even with the main screed.

NOTE: Do not fill in the area directly in front of the extender on rear-mounted extender screeds. This area will be filled by material automatically as the paver pulls forward off the starting reference.



STEP 13

Step 13 is setting all feeder controls.

When you have filled the auger chamber half full—no more, no less—put the conveyor and auger mode switches on the operator's console to Auto.